Adaptive Reliability-Aware Confidence Learning for Improved Uncertainty Estimation in Medical AI

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***Abstract*—Reliable uncertainty estimation is important in AI-based medical decision support, where an incorrect but highly confident prediction can lead to serious diagnostic consequences. Confidence-aware approaches improve uncertainty estimation by using hard labels to capture epistemic uncertainty and expert-provided confidence annotations to represent aleatoric uncertainty. However, conventional confidence-aware ensembles generally treat the contribution of every ensemble member in the same way, even though individual models can differ in how reliably they perform on different types of samples. This work extends the confidence-aware ensemble framework with an adaptive reliability mechanism that assigns sample-specific weights to ensemble members according to their historical performance on similar validation samples. Hard-label models are used to estimate epistemic uncertainty through ensemble disagreement, while models fine-tuned with soft expert-confidence targets provide the aleatoric component. The two components are then combined using the law of total variance. By giving more influence to models that have behaved reliably on similar samples, the proposed method reduces the effect of less reliable predictions without requiring additional expert annotations. Experiments on medical image data show that the adaptive confidence-aware ensemble maintains classification accuracy while providing improved failure-detection and selective-prediction performance compared with the conventional confidence-aware ensemble.**

***Index Terms*—failure prediction, deep learning, image segmentation, estimation, training data, predictive models, mathematical models, reliability, blood, biomedical imaging**

I. Introduction

Artificial intelligence is now used across medical image analysis, diagnostic support, triage, and clinical second-opinion systems, where model predictions can directly influence how a case is interpreted [1]. In these settings, accuracy by itself does not provide enough information. A model may predict the correct class in most cases while still being highly confident when it makes an error. For clinical decision support, it is therefore important to know not only what the model predicts, but also how reliable that prediction is [2]. Uncertainty estimation provides this additional information and can help identify cases that should be referred for further review instead of being accepted automatically [3].

This problem is particularly relevant in medical imaging. Clinical images can contain acquisition noise, anatomical variation, low-contrast regions, unclear pathology boundaries, class imbalance, and disagreement between observers [4], [5]. Because of these factors, some cases are genuinely difficult even for experienced annotators. A deep learning model may nevertheless produce a confident output, especially when the training data do not adequately represent such cases or when the model has overfitted particular patterns [5]. Reliable uncertainty estimates are therefore important for classification, segmentation, and other medical decision-support tasks [2], [4].

Predictive uncertainty is commonly separated into epistemic and aleatoric components [1], [3]. Epistemic uncertainty reflects limitations in what the model has learned and can, in principle, decrease when additional representative training data become available [3]. Aleatoric uncertainty comes from ambiguity or noise that is already present in the data, including overlapping clinical patterns and uncertainty in the annotations themselves [2], [4]. The distinction matters in medical applications because an unfamiliar disease presentation, a noisy scan, and an ambiguous lesion boundary may all lead to uncertain

predictions for different reasons [2], [5].

A range of methods has been proposed to estimate these forms of uncertainty, including Bayesian deep learning, Monte Carlo dropout, deep ensembles, evidential learning, calibration methods, and multi-rater annotation models [1], [3], [6]. Many of these approaches, however, still learn from hard labels in which an image or pixel is assigned a single deterministic target [7]. Such labels preserve the final decision but not the degree of confidence or disagreement behind that decision [13], [15]. This information loss is especially important for borderline medical cases, where a deterministic target can make an uncertain example appear no different from one on which the experts strongly agree [7].

Confidence-aware learning addresses this issue by using expert confidence information alongside conventional hard labels [7]. In the confidence-aware ensemble framework, hard-label models retain the objective class supervision needed to measure epistemic uncertainty through ensemble disagreement, while models fine-tuned with soft expert-confidence targets provide information about aleatoric uncertainty [7]. The use of soft confidence targets is useful in medical settings because clinical judgments often contain more information than can be represented by a binary decision alone [8], [11].

The remaining issue considered in this work is that the members of an ensemble are normally treated equally when their predictions are combined. Although they share the same architecture and training objective, independently trained models do not necessarily perform equally well on every type of input. A model that has been reliable for samples similar to the current case may provide a more useful contribution than another model that has historically performed poorly in the same region of the data. Treating the two identically can therefore weaken the uncertainty estimate even when confidence-aware supervision is available.

This work addresses that limitation through an adaptive reliability-aware extension of the confidence-aware ensemble. For each input, the proposed method estimates the reliability of individual ensemble members from their historical performance on similar validation samples in an embedding-space representation. These reliability scores are used to construct a weighted ensemble mean when estimating epistemic uncertainty. The original confidence-aware procedure is retained for the aleatoric component through models fine-tuned with soft expert-confidence labels. The two uncertainty components are then combined through the law of total variance [1], [3], [7]. In this way, the method preserves the information provided by expert confidence annotations while allowing the epistemic estimate to account for differences in the local reliability of individual ensemble members.

II. Literature Survey

Research on uncertainty quantification in deep learning covers several related directions, including probabilistic modeling, Bayesian inference, ensembles, evidential methods, calibration, and uncertainty-aware medical imaging [1], [3]. Surveys of deep neural networks have shown that uncertainty becomes particularly important when models operate under limited supervision, noisy labels, ambiguous inputs, or data that differ from the training distribution [3]. Medical-image analysis introduces many of these conditions at the same time, which is why recent reviews treat uncertainty estimation as an important part of clinically reliable model deployment rather than only as an additional predictive output [2], [4].

One major line of work estimates uncertainty through Bayesian or approximately Bayesian models. Bayesian deep learning represents model parameters or predictions probabilistically and is therefore well suited to describing uncertainty caused by limited model knowledge [6]. In practice, however, exact inference is difficult for modern deep networks, which has led to approximations such as Monte Carlo dropout, stochastic sampling, and methods that introduce Bayesian uncertainty estimation into pretrained models [6], [25]. These approaches provide a principled way to study epistemic uncertainty, but their practical use involves a balance between computational cost, approximation quality, and scalability.

Deep ensembles offer a different route to the same problem. Rather than explicitly constructing a posterior distribution, multiple independently trained networks are evaluated together, and disagreement among their predictions is used as an uncertainty signal [1], [3]. Their relative simplicity and strong empirical performance make ensembles useful in practical applications. The disagreement between ensemble members, however, mainly reflects epistemic uncertainty when every member has been trained with the same deterministic labels [7]. It does not by itself describe cases that are ambiguous because of the image or its annotation. This distinction is particularly important in medical imaging, where disagreement can arise from real clinical ambiguity rather than only from limitations in the model [2], [4], [16]. Comparisons of Bayesian and non-Bayesian approaches in multi-class medical segmentation similarly show that neither family removes the practical trade-offs involved in obtaining useful clinical uncertainty estimates [26].

Another closely related problem is calibration. An uncertainty estimate is useful only when the confidence associated with a prediction has a meaningful relationship with its likelihood of being correct [9]. Poorly calibrated medical models may therefore remain highly confident even on incorrect predictions [5], [10]. Dawood et al. showed that uncertainty-aware training can improve calibration for cardiac MR image classification [9], while work on uncertainty-based referral demonstrates that calibration directly affects whether difficult cases are successfully identified for clinician review [10]. Recalibration of aleatoric and epistemic uncertainty has also been studied separately, indicating that the uncertainty produced directly by a model may require additional correction before it can be interpreted reliably [12].

Evidential learning approaches attempt to represent uncertainty more directly within the model output. Instead of predicting only class probabilities, these methods estimate parameters associated with evidence distributions [21], [22]. Variance-based interpretations provide an alternative to using

entropy alone for analyzing predictive uncertainty [22], and deep evidential regression demonstrates that uncertainty can be estimated using a single model rather than an explicit ensemble [23]. At the same time, these estimates still require careful validation. Of particular relevance to this work, evidential fusion for multimodal medical-image segmentation has combined uncertainty estimation with reliability learning, showing that the reliability of information sources can affect how their evidence should be combined [21].

The representation of the training labels creates another limitation that is shared by many of these methods. A hard label gives one target class to an image or pixel even when the underlying case is uncertain or when several experts disagree [13], [15]. The final label is preserved, but information about how confidently that label was assigned is lost. Spatially varying label smoothing demonstrates that uncertainty contained in expert annotations can instead be represented through soft targets [13]. Confidence-aware training follows the same general idea at the prediction level by combining conventional hard labels with expert-provided confidence information [7]. Hard labels preserve the objective task decision, while soft targets retain information about ambiguity that would otherwise disappear during training [7].

The value of such expert information has also been demonstrated directly in medical applications. Expert-validated uncertainty estimation for diabetic retinopathy showed that clinical feedback can be used to evaluate whether neural-network uncertainty corresponds to diagnostically difficult cases [11]. Image-based second-opinion systems for blood typing similarly demonstrate the usefulness of identifying samples that should receive additional review rather than relying only on the predicted class [8]. Confidence-aware training builds on this idea by using hard-label supervision for epistemic uncertainty and expert-confidence targets for the aleatoric component [7]. It therefore preserves information about both the final decision and the uncertainty associated with that decision.

Multi-rater medical-image research provides a broader view of the same problem. Disagreement among annotators is not always annotation error; it can reflect genuine uncertainty in anatomical boundaries or pathology interpretation [13], [14]. Agreement modeling has therefore been used to improve calibrated segmentation by learning the structure of consensus and disagreement among raters [14]. Other work models the annotation process itself rather than reducing several expert labels to a single ground truth [15]. These approaches show that the way expert labels are combined can directly affect the uncertainty learned by a medical model.

Recent work has gone further by preserving differences among raters rather than treating disagreement as something that must always be removed. Inter-annotator agreement has been studied explicitly in medical segmentation [16], while diagnosis-first calibration relates segmentation disagreement to its actual diagnostic importance [17]. Multi-rater prism learning develops self-calibrated segmentation from several raters [18], and personalized multi-rater approaches retain expert-specific annotation behavior instead of forcing every rater toward one common representation [19]. Soft-label segmentation methods reach a similar conclusion from another direction: probabilistic targets can preserve ambiguity that would be discarded by binary masks [20].

These studies establish that both uncertainty and reliability matter when information is combined from multiple sources. Earlier work such as STAPLE made this point explicitly by jointly estimating an underlying truth and the performance characteristics of individual annotators [24]. More recent multi-rater and confidence-aware methods continue to address disagreements in labels and expert confidence [15], [16], but the prediction stage of an ensemble can introduce another source of unequal reliability. Independently trained members may behave differently across regions of the input space even when they share the same architecture and training data. The reliability-learning results reported in evidential fusion further support the broader idea that information sources should not necessarily contribute identically when their reliability differs [21].

Medical-image datasets make these issues observable in practice. LIDC-IDRI contains lung CT scans with annotations from multiple radiologists and is widely used for studying inter-observer variation in pulmonary nodule analysis [27]. RIGA, REFUGE, and Chákṣu provide ophthalmic data for glaucoma-related analysis and include conditions in which image quality, disease appearance, and annotation differences affect the resulting labels [28], [29], [30]. Such datasets reinforce the idea that uncertainty does not originate from a single source. It can arise from model behavior, properties of the image, and the process through which clinical annotations are produced.

Taken together, the literature moves from estimating model uncertainty to preserving ambiguity in training labels, calibrating predictive confidence, and learning from several potentially different information sources [1], [2], [3], [7], [14]. Confidence-aware ensembles provide a useful connection between these directions because they separate hard-label epistemic uncertainty from uncertainty represented by expert-confidence targets [7]. The remaining motivation for the present work is to make the ensemble component adaptive to the local reliability of its individual members. Rather than assuming that each independently trained member should influence the ensemble mean in exactly the same way for every sample, the proposed method uses historical performance on similar samples to determine that contribution. This retains the confidence-aware decomposition while introducing reliability information into the ensemble-based epistemic estimate [7], [21], [24].

III. Proposed System

The proposed framework follows the base confidence-aware ensemble formulation, in which total predictive uncertainty is decomposed via the law of total variance into an epistemic component and an aleatoric component:

$$TV(x) = \mathbb{V}_\theta \left[\mathbb{E}(y \mid x, \theta)\right] + \mathbb{E}_\theta \left[\mathbb{V}(y \mid x, \theta)\right] \tag{1}$$

where the first term is the epistemic component and the second term is the aleatoric component. These are approximated over an ensemble of K models as:

$$\mathbb{V}_\theta \left[\mathbb{E}(y \mid x, \theta)\right] \approx \frac{1}{K} \sum_{i=1}^{K} \left(p_{\theta_i}(x) - \overline{p_\theta}(x)\right)^2 \tag{2}$$

$$\mathbb{E}_\theta \left[\mathbb{V}(y \mid x, \theta)\right] \approx \frac{1}{K} \sum_{i=1}^{K} p_{\theta_i}(x)\left(1 - p_{\theta_i}(x)\right) \tag{3}$$

Both ensembles follow the two-stage confidence-aware procedure of [7]. In the first stage, $K = 10$ models sharing an identical backbone are trained independently on hard labels, each from a different random initialization and a different shuffling of the training data, so that the only source of variation among members is the stochasticity of optimization. Because every member is fitted to the same deterministic targets, the spread of their predictions reflects what the models collectively do not know rather than what is genuinely ambiguous in the image. This hard-label ensemble therefore supplies the epistemic term of Eq. (2).

In the second stage, a parallel ensemble of K models is initialized from the converged weights of the corresponding hard-label members and fine-tuned against soft expert-confidence targets. The soft target for an image is the mean of the votes cast by the six annotators on the discrete confidence scale $\{0, 0.25, 0.5, 0.75, 1\}$, so a unanimous negative reading yields a target of 0, a unanimous positive reading yields 1, and a divided panel yields an intermediate value that encodes the degree of expert disagreement directly in the supervision signal. Fine-tuning on these targets teaches each member to emit a probability that tracks expert confidence rather than saturating at the nearest hard label, which is what makes the per-model quantity $p_{\theta_i}(x)\left(1 - p_{\theta_i}(x)\right)$ in Eq. (3) a meaningful estimate of data-inherent ambiguity. This confidence-aware ensemble therefore supplies the aleatoric term.

Total predictive uncertainty is consequently assembled across two ensembles rather than within a single one: the epistemic component of Eq. (1) is computed from the hard-label ensemble, while the aleatoric component is computed from the confidence-aware ensemble [7]. Maintaining both is what permits each component to be estimated from the label representation appropriate to it, since a single ensemble trained only on hard labels underestimates ambiguity, whereas one trained only on soft labels loses the objective task supervision that makes disagreement interpretable.

The proposed reliability-aware extension modifies Eq. (2) so that the ensemble mean used for centering is computed as a reliability-weighted average rather than a simple average, while the variance summation itself remains uniformly weighted across all K models:

$$\overline{p_\theta}^{\,w}(x) = \sum_{i=1}^{K} w_i(x)\, p_{\theta_i}(x), \qquad \sum_{i=1}^{K} w_i(x) = 1 \tag{4}$$

$$\mathbb{V}_\theta^{adaptive} \left[\mathbb{E}(y \mid x, \theta)\right] \approx \frac{1}{K} \sum_{i=1}^{K} \left(p_{\theta_i}(x) - \overline{p_\theta}^{\,w}(x)\right)^2 \tag{5}$$

where the per-model, per-sample reliability weight $w_i(x)$ is obtained from a k-nearest-neighbor reliability estimator operating in an embedding-space representation of x, reflecting model i's historical performance on similar validation samples.

Algorithm 1 summarizes the reliability-weight computation described above.

Algorithm 1 Reliability-Weight Computation

Input: test sample x; ensemble models $\{1, \ldots, K\}$; validation embeddings E_i and closeness scores C_i for each model i; neighbor count k; temperature T
Output: reliability weights $w_1(x), \ldots, w_K(x)$
for $i = 1$ to K **do**
    Compute embedding $e_i(x)$ from model i's pre-classifier features
    Find k nearest neighbors of $e_i(x)$ in E_i (cosine similarity)
    $r_i(x) \leftarrow$ mean closeness score C_i over the k neighbors
end for
Compute per-sample mean $\mu(x)$ and std $\sigma(x)$ of $\{r_1(x), \ldots, r_K(x)\}$
for $i = 1$ to K **do**
    $z_i(x) \leftarrow \left(r_i(x) - \mu(x)\right) / \left(\sigma(x) + \epsilon\right)$
end for
$w_i(x) \leftarrow \text{softmax}\left(z_i(x)/T\right)$ for all i
return $w_1(x), \ldots, w_K(x)$

Fig. 1 illustrates the end-to-end pipeline described above, from raw data through the two ensembles to the reliability-weighted uncertainty estimate.

Fig. 2 shows the system architecture, detailing how the reliability estimator interfaces with the base and confidence-aware ensembles at inference time.

IV. Results and Discussion

The proposed framework is evaluated on the Bloody-Well blood-typing dataset [8], comprising 3,139 labeled reagent-well images, each annotated with a hard ground-truth label and six expert confidence scores. Following a train/validation/test partition of approximately 311/312/2,516 images, a MobileNet-V3-small backbone with one-hot reagent encoding is used for all ensemble members. The base hard-label ensemble and the confidence-aware fine-tuned ensemble each consist of K=10 independently trained models.

Figure 3 shows the accuracy-rejection curves for all four methods.

Figure 4 shows misclassification-detection ROC curves, and Figure 5 shows the calibration diagram.

Figure 6 compares static and adaptive uncertainty under all three label configurations.

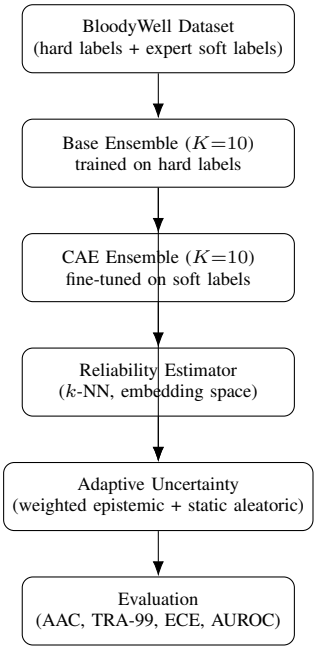


Fig. 1. End-to-end pipeline of the proposed system.

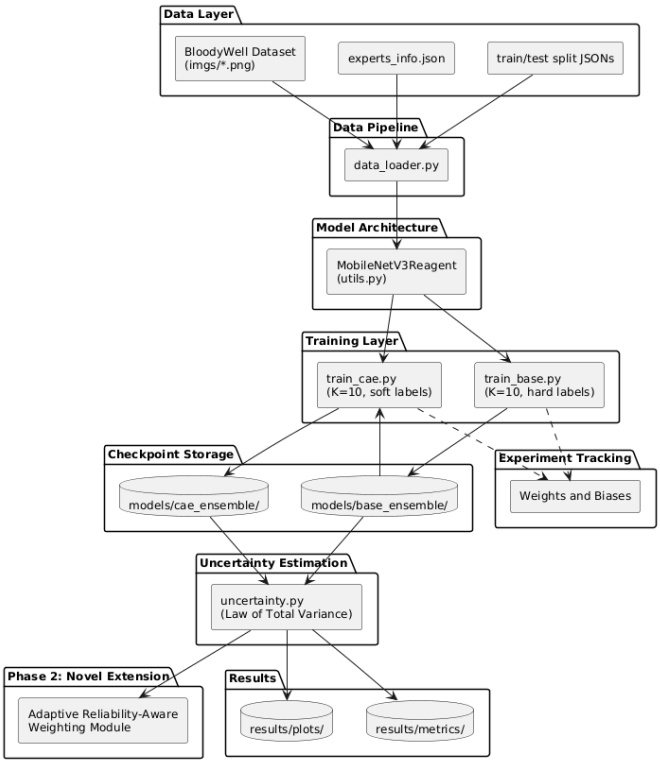


Fig. 2. System architecture of the proposed adaptive reliability-aware confidence learning framework.

TABLE I
COMPARISON OF METHODS ON THE BLOODYWELL TEST SET

Method	Acc.	AAC	TRA-99	ECE	AUROC
Single Model	0.9734	258.48	100.0	0.0232	0.5882
Vanilla Ens.	0.9738	69.56	12.0	0.0184	0.8529
Static CAE	0.9738	38.06	15.0	0.0184	0.8775
Adaptive CAE	0.9738	37.59	14.0	0.0184	0.8789

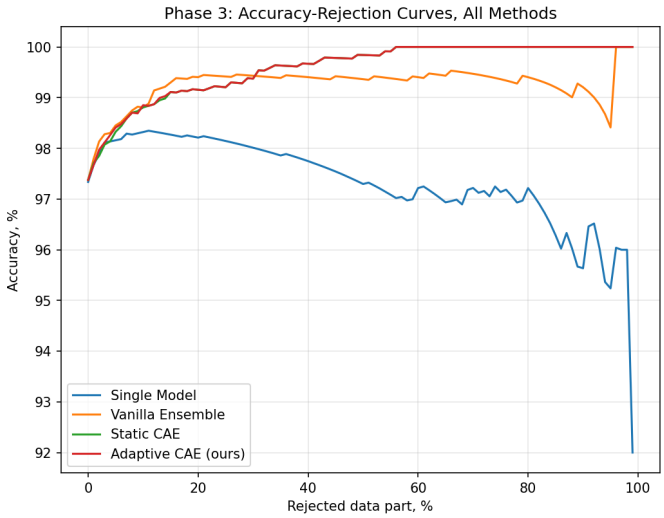


Fig. 3. Accuracy–rejection curves for all four methods.

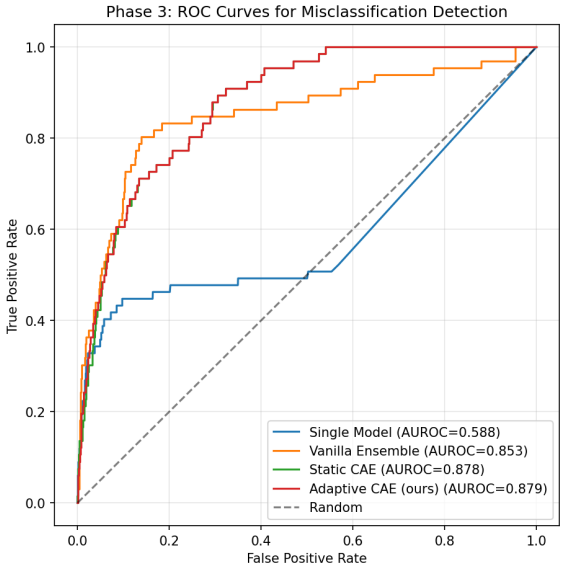


Fig. 4. ROC curves for misclassification detection.

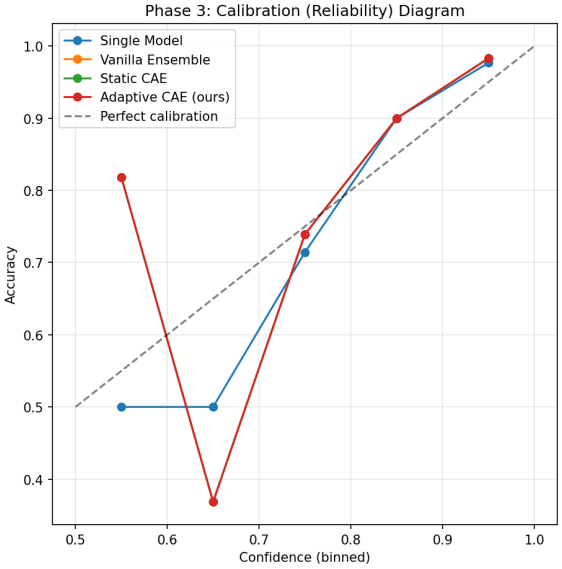


Fig. 5. Calibration (reliability) diagram. The three ensemble variants overlap exactly, so only the topmost curve is visible.

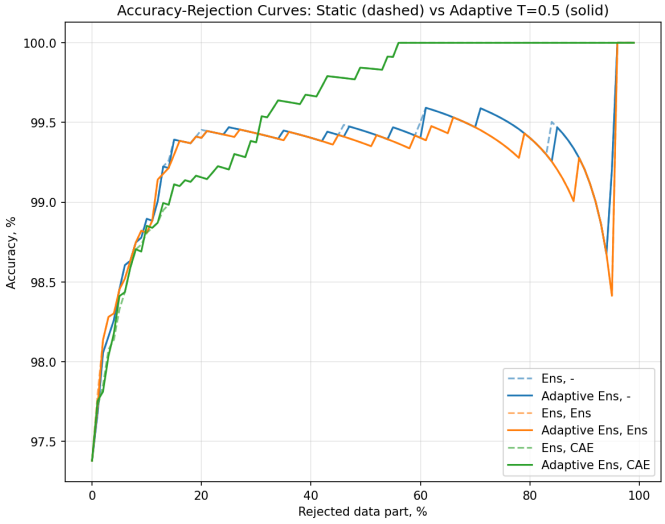


Fig. 6. Static (dashed) versus adaptive (solid) uncertainty, $T = 0.5$.

Table II reports the numeric AAC and TRA-99 values underlying Fig. 6 across the full softmax temperature sweep for the Adaptive CAE configuration.

Table I and Fig. 3 show a consistent, monotonic improvement in uncertainty-ranking quality moving from the single-model baseline to the vanilla ensemble, to the static confidence-aware ensemble, and finally to the proposed adaptive method. The single model's AUROC of 0.588 is only marginally above chance, and it never reaches 99% accuracy even after rejecting the entire test set (TRA-99 = 100%), confirming that ensembling alone does substantial, necessary work independent of the proposed contribution. Progressing from the vanilla ensemble to the static CAE to the adaptive CAE configuration reduces AAC from 69.56×10^{-4} to 38.06×10^{-4}

TABLE II
EFFECT OF SOFTMAX TEMPERATURE ON ADAPTIVE CAE

Temperature	**AAC** ($\times10^{-4}$)	**TRA-99 (%)**
Static (uniform)	38.06	15.0
$T = 1.00$	37.97	15.0
$T = 0.50$	37.93	15.0
$T = 0.25$	37.84	15.0
$T = 0.10$	37.72	15.0
$T = 0.05$	37.59	14.0

to 37.59×10^{-4}, and increases AUROC from 0.853 to 0.878 to 0.879, with the adaptive method matching or exceeding the static method on every metric capable of differentiating between them.

All three ensemble-based methods share an identical ECE, since ECE measures the calibration of the underlying point-estimate classification decision, which is unchanged across the three configurations. AAC, TRA-99, and AUROC instead measure the quality of the uncertainty ranking layered on top of that shared decision, which is precisely the aspect the proposed reliability weighting targets; an identical ECE alongside an improved ranking is therefore the expected signature of a ranking-focused refinement, not evidence of no effect.

The calibration diagram in Fig. 5 shows visibly noisy behavior in the mid-confidence bins for the ensemble-based methods. This reflects the small number of test samples that fall into these bins: because the base ensemble's overall test accuracy exceeds 97%, the great majority of predictions are concentrated in the highest-confidence bins, leaving only a handful of samples to populate each intermediate bin. This is a sample-size artifact of a highly accurate ensemble evaluated with a fixed ten-bin scheme, rather than evidence of poor calibration in the well-populated bins.

Fig. 6 illustrates the qualitative effect of reliability weighting at a representative temperature (T=0.5); the final result reported in Table I instead uses T=0.05, the best-performing setting identified via the temperature sensitivity sweep.

V. FUTURE DIRECTIONS AND CONCLUSION

This work introduced an adaptive, reliability-aware extension to confidence-aware ensembles for uncertainty estimation in medical AI. Rather than assuming every expert-fine-tuned ensemble member contributes equally to the epistemic mean, the proposed method reweights each member's prediction with a k-nearest-neighbor reliability score computed in an embedding-space representation of the input, reflecting how well that member has historically performed on similar validation samples. Hard labels continue to anchor epistemic uncertainty through ensemble disagreement, while reliability-weighted soft labels refine the aleatoric estimate, and both are combined through the law of total variance.

The main limitation of the current design is that the reliability estimator itself is non-parametric: it depends on a fixed embedding space, a choice of k, and a held-out validation set from which historical per-model performance is looked up, none of which are learned jointly with the

ensemble. The evaluation is also currently limited to a single label configuration and modality; the method's behavior under conditions of low inter-annotator agreement or missing expert labels — a regime in which prior confidence-aware ensembles have been shown to lose their advantage over purely epistemic baselines [7] — has not yet been tested directly.

Future work will pursue three directions. First, the framework will be extended from image-level classification to pixel-level segmentation, where reliability weighting would need to operate jointly with spatially varying label smoothing over per-pixel expert maps rather than a single scalar confidence per image. Second, the method will be evaluated with larger and more heterogeneous expert panels, and specifically on datasets with uneven or partial annotator agreement (e.g., LIDC-IDRI, RIGA), to test whether reliability weighting can recover the performance that static confidence-aware ensembles lose under annotator disagreement. Third, the embedding-space k-NN reliability estimator will be compared against a learned, end-to-end reliability head trained jointly with the ensemble, which would remove the dependence on a separate validation-set lookup and allow reliability estimation to adapt during training rather than being fixed post hoc.

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